PRANAV GOYAL

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EDUCATION

University of Michigan, Ann Arbor *Aug 2024 - Aug 2029*
Ph.D. in Robotics, Advisor: Dr. Christoforos Mavrogiannis *GPA 4.12/4.00*

BITS Pilani, K.K. Birla Goa Campus *Aug 2019 - Mar 2023*
B.E. in Electronics and Instrumentation Engineering *GPA 8.58/10*

RESEARCH EXPERIENCE

Fluent Robotics Lab, University of Michigan Ann Arbor, USA
Graduate Research Assistant *Aug 2024 - Present*

- Built MPPI-based intent-expression planners in PyTorch and ran a study on a Hello Robot Stretch 2 showing that distraction changes what makes robot motion legible (HRI 2026; THRI 2026).
- Co-led Bi3 and ACME, social navigation datasets spanning two platforms and two cultures, for benchmarking foundational navigation models (ICRA 2026; under review at IJRR).
- Developing Alpaca, an open ROS 2 platform for interactive social navigation (under review, HRI 2027).

CLeAR Lab, National University of Singapore Singapore
Research Assistant, Advisor: Dr. Harold Soh *Aug 2023 - Aug 2024*

- Formulated LTLDoG, a diffusion-based global planner satisfying temporal-logic constraints, deployed on a Unitree Go2 (RA-L 2024).
- Designed SocRATES, an LLM-powered tool turning natural-language scenarios into executable ROS2/Gazebo benchmarks for social navigation (RA-L 2026; Runner-up Best Paper, RSS 2024 workshop).

Biorobotics Lab, Robotics Institute, CMU Pittsburgh, USA
Research Intern, Advisor: Dr. Howie Choset *Jun 2022 - Jan 2023*

- Implemented convex MPC and geometric control for a legged-wheeled robot and built its prototype from HEBI modules (DARPA VeRT; Bachelor's thesis).
- Built an automated void detection pipeline from point cloud registration and SfM reconstruction of drone footage of the Surfside collapse (IROS 2023, SSRR 2022).

SELECTED PUBLICATIONS

- **Rethinking Legibility in Social Robot Hallway Navigation.** **Goyal** et al. *THRI 2026.*
- **Alpaca: An Interactive Platform for Social Robot Navigation.** **Goyal** et al. *Under review, HRI 2027.*
- **How Human Motion Prediction Quality Shapes Social Navigation.** Stratton et al. *HRI 2026.*
- **Bi3: A Biplatform, Bicultural, Biperson Social Navigation Dataset.** Stratton et al. *ICRA 2026.*
- **SocRATES: Scenario-based Testing of Social Navigation.** Marpally, **Goyal**, Soh. *RA-L 2026.*
- **ACME: Multi-Cultural, Multi-Embodiment Navigation Dataset.** Marpally et al. *Under review, IJRR.*
- **LTLDoG: Symbolic Constraints for Safe Diffusion Planning.** Feng, Luan, **Goyal**, Soh. *RA-L 2024.*
- **Void Detection for Search and Rescue with 3D Perception.** Bal, **Goyal**, et al. *IROS 2023.*

SKILLS

Languages & Tools: Python, C/C++, MATLAB, R, Bash; PyTorch, ROS 1/2, Gazebo, OpenCV, Open3D
Methods: Optimal and sampling-based control (MPPI, convex MPC), diffusion planning, reinforcement learning, human motion prediction, 3D reconstruction (SfM, point cloud registration), human detection and tracking
Robot Platforms: Hello Robot Stretch 2, Unitree Go2, Universal Robots and WidowX arms, AgileX Ranger Mini 3.0, OptiTrack

AWARDS & SERVICE

Awards: Robotics PhD Departmental Fellowship | Rackham Travel Grant | Runner-up Best Paper, RSS 2024 Social Navigation Workshop
Reviewer: THRI, RA-L, ICRA, IROS, HRI, Autonomous Robots, RO-MAN
Teaching: Graduate Student Instructor, ROB 204: Introduction to Human-Robot Systems (Fall 2026)